



EDPACS

The EDP Audit, Control, and Security Newsletter

ISSN: 0736-6981 (Print) 1936-1009 (Online) Journal homepage: www.tandfonline.com/journals/uedp20

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To cite this article: Abdeslem Blali, Souhayla Dargaoui, Mourade Azrou, Azidine Guezzaz, Fatima Amounas & Abdulatif Alabdulatif (2025) Analysis of deep learning-based intrusion detection systems in IoT environments, EDPACS, 70:11, 18-52, DOI: [10.1080/07366981.2025.2498222](https://doi.org/10.1080/07366981.2025.2498222)

To link to this article: <https://doi.org/10.1080/07366981.2025.2498222>



Published online: 06 May 2025.



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ANALYSIS OF DEEP LEARNING-BASED INTRUSION DETECTION SYSTEMS IN IOT ENVIRONMENTS

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Abstract. The present paper looks at the security problems caused by the fast growth of the Internet of Things in areas like industry, healthcare, and agriculture. As internet of things systems become more common, they face more threats from cyberattacks. To deal with these security issues, we studied recent papers to review different intrusion detection systems made for the Internet of Things. The goal was to see how well they work and find ways to make them better. Our research shows that deep learning-based intrusion detection systems can improve how we detect online threats. These systems, especially when they use neural networks, are better at spotting and reacting to harmful activities. Combining machine learning with intrusion detection systems seems to help boost the security of Internet of Things networks, offering stronger protection against cyberattacks. One of the best algorithms we found was the combination of a convolutional neural network and a long short-term memory network. This deep learning model showed very high accuracy in protecting Internet of Things networks, especially when tested with different datasets. This proves that using advanced algorithms is important to keep up with the growing challenges of cyber threats targeting IoT systems.

1. INTRODUCTION

The Internet of Things (IoT), projected to exceed 30 billion active device connections globally by 2025, presents an expansive attack surface. In fact, IoT technologies have changed the way devices interconnect and share data (Almaraz-Rivera et al., 2023; Tawalbeh et al., 2020). This transformation has provided significant benefits in various fields like healthcare, smart home, and agriculture. However, it brings with it different security issues, such as data violations, non-authorized access, and system viabilities. To address these challenges, strong security solutions are required. Hence, key security features include encrypting data to block unauthorized access, authorizing devices to guarantee their legitimacy, and keeping software regularly updated to correct any vulnerability. These strategies are vital for protecting IoT devices and ensuring the stability and safety of interconnected systems (Zhang et al., 2024).

The first important solution is data encryption. When IoT devices communicate with each other, they often exchange sensitive information. Encryption makes this information unreadable to unauthorized people by converting it into a complex code. This system uses encryption keys that are needed to decrypt the data. So, even if someone intercepts the information, it remains protected and impossible to understand (Odyuo et al., 2023).

Device authentication and identity management are also essential in an IoT network. Each device must prove it is authorized to access the network. This can be done with complex passwords or more advanced systems like multi-factor authentication. These algorithms ensure that only legitimate users and devices can interact with the network, reducing the risk of intrusions by malicious individuals (Isong et al., 2024).

Finally, regular software updates are another important solution for the security of IoT devices. IoT devices can have vulnerabilities in their programs that hackers can exploit. Additionally, Intrusion Detection Systems (IDS) play a crucial role in IoT security by monitoring network traffic for signs of malicious activity. IDS can quickly detect and alert users to potential cyberattacks, providing an extra layer of defense that complements software updates and other security measures (Liao et al., 2024). IDS play a key role in protecting against different risks (Almaraz-Rivera et al., 2023). However, the traditional methods used for designing IDSs face major issues in various IoT networks. However, the fact that they are not sufficiently adaptable to the variety of different IoT technologies and protocols decreases the efficiency

In the past years, researchers have used both Machine Learning (ML) and Deep Learning (DL) algorithms to improve IDS efficiency (Aldhaheeri et al., 2024; Alsoufi et al., 2021; Aversano et al., 2021; Ghaffari et al., 2024; Hatcher & Yu, 2018; Hussain & Bhatti, 2024; Muneer et al., 2024; Rafique et al., 2024; Sharma et al., 2024; Thomas & Bhat, 2021). Hence, ML allows the systems to automatically gather valuable information from large amounts of data. Besides, with sufficient training dataset, ML-based IDS can successfully detect threats. Moreover, these models can be built and use without investigating hard efforts. On the other hand, DL uses models that are designed to work like the human brain's neural networks. These models help solve problems related to IoT devices by understanding data and applying it in various fields. In addition, DL algorithms are generally more effective than ML, especially when dealing with large datasets. However, IDS still face challenges in quickly detecting intrusions because of the growing network traffic and security risks.

This paper focuses on reviewing recent papers that address IDS using deep learning and explores how these systems can improve the security of IoT devices. It also looks at the limitations of each system. The main research questions include the algorithms used

in the recent years, the deep learning algorithms chosen, the complexity of each proposal, and the types of attacks included in the datasets used to develop deep learning-based IDS.

The rest of the article is as follows: Section “Materials and Methods” discusses the security challenges facing the IoT ecosystem. Section “Intrusion Detection Systems Employed in the IoT Environment” examines the security solutions implemented to address these challenges. Section “Conclusion” reviews related work to provide context for ongoing research. The fifth section focuses on intrusion detection systems deployed in IoT environments, detailing the techniques and algorithms used. Section “References” concludes with a summary of the findings and possible future directions for improving IoT security.

MATERIALS AND METHODS

This section describes the comprehensive methodology used to conduct the systematic review of Deep Learning-based intrusion detection approaches adopted recently in IoT systems. The methodology followed is based on the guidelines described in the PRISMA method.

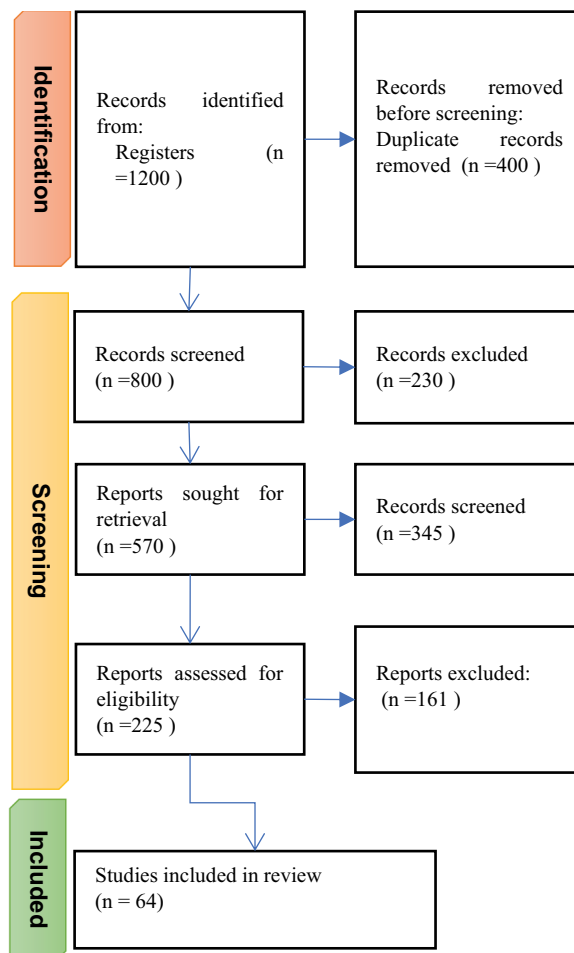
Search strategy

Our study was carried out over the period 2020–2024, as IoT IDS has recently received significant attention. Research conducted over the past five years presents emerging technologies and current trends in DL-based IDS effectiveness, helping to strengthen understanding of the state of the art. In order to identify the paper that we can adopt in our analysis, we have done research in various databases like: Google Scholar (<https://scholar.google.com/>), ACM Digital Library (<http://dl.acm.org>), IEEE eXplore (<http://ieeexplore.ieee.org>), and ScienceDirect (<https://www.sciencedirect.com>). In this research, we have determined the specific search words to find the appropriate papers. Hence, three fields were carefully examined in our search which title, abstract, and keywords. The search expressions were created based on numerous keywords conforming to the suitable conditions and combined with Boolean operators (AND, OR, NOT). The search expressions used are:

- IoT: “IoT”, “Internet of Things”, “IoT system”
- Intrusion detection: “intrusion detection”, “anomaly detection”
- Deep learning: “artificial intelligence”, “deep learning”, “Neural Network”.

Study selection

After gathering various papers found in queries step, we have done the selection of the paper we will analyze. The main goal of this step is to eliminate studies that are not relevant to our analysis. Hence, this goal is accomplished through a filtering process

Figure 1 *PRISMA flowchart.*

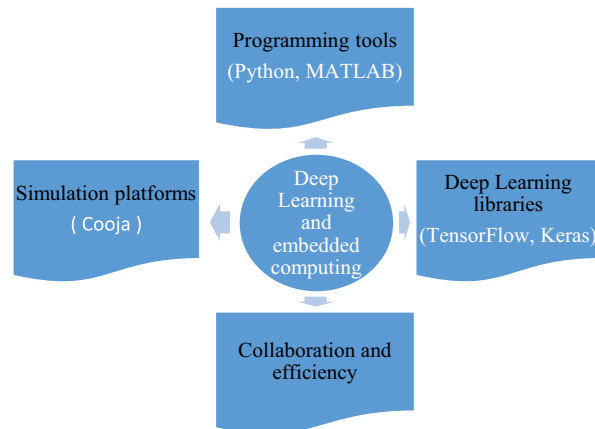
that usually begins with the verification of study titles, abstracts, and whole texts. Once this pre-selection has been completed, we can obtain the remaining studies to assess their relevance. Hereafter, the complete flowchart of the selection process, comprising identification, selection, qualification, and inclusion, is illustrated in [Figure 1](#).

INTRUSION DETECTION SYSTEMS EMPLOYED IN THE IOT ENVIRONMENT

Analysis of adopted software

The tools and technologies used in deep learning and embedded computing are very important because they help researchers and developers create and improve innovative projects ([Figure 2](#)). Tools like Python, MATLAB, and simulation platforms such as Cooja provide environments for programming, modeling, and testing ideas. Libraries like TensorFlow and Keras make it easier to handle complex tasks like building and training deep learning

Figure 2 Key tools and their roles in deep learning and embedded computing.



models. These tools save time, encourage teamwork, and provide a common way for researchers to work. Understanding their role helps us see how they influence research and practical applications in these fields (Wang et al., 2024).

An organized view of the various technologies and software associated with deep learning and embedded computing is presented. It highlights the tools used in creating and implementing projects in these fields, including development environments such as Python, MATLAB, and Cooja, as well as widely used libraries like TensorFlow, Keras, and Scikit. The inclusion of checkmarks indicates the application of these tools in specific studies, offering a clear representation of the technological choices made during different research or development stages.

The preferences and orientations of developers and researchers in their exploration of artificial intelligence and embedded computing are revealed (Khacha et al., 2023). The intersections of technologies such as Python, TensorFlow, and Keras suggest a combined use of these popular tools to create deep learning models. Similarly, the inclusion of Raspberry Pi and Cooja highlights the interest in embedded solutions and experimentation in real or simulated environments.

Python is the most used programming language in the reviewed papers. It appears 16 times, representing 50% of the analyzed papers (Kilichev et al., 2024). This frequent use suggests that Python is often paired with frameworks and platforms like Keras or TensorFlow. In contrast, MATLAB is used less frequently, appearing in only three papers.

Finally, it is clear that researchers prefer to conduct tests and simulations with simulators like Cooja. It is rare for them to deploy their frameworks on embedded systems such as the Raspberry Pi, which is mentioned only in the study (Awajan, 2023).

Analysis of the used datasets

Deep learning-based IDS require datasets to evaluate intrusions effectively. Creating suitable data for model training is crucial yet challenging, as it involves identifying labeled normal and abnormal communications and other features such as IP addresses. However, some datasets used for network traffic analysis remain unavailable to the public due to security concerns. Various datasets utilized in the reviewed studies are described, providing insights into their characteristics and applications (Khacha et al., 2023).

As it is depicted in Table 1, the datasets used for intrusion detection show considerable diversity in terms of IoT specificity, publication years, characteristics, classes, and volumes of normal and attack records. Notable datasets include TON-IOT (2020) with 44 features and 161,043 attack records, and BoT-IoT (2018) with 46 features, 5 classes, 477 normal records, and 3,668,045 attack records. Other widely used datasets like NSL-KDD (1998) and UNSW-NB15 (2015) are not specific to IoT but remain popular due to their rich feature sets and classes.

Table 1 Datasets used for intrusion detection.

Dataset	Year of publication	Features	Number of classes	Total normal records	Total attack records
TON-IOT	2020	44	-	-	161043
NSL-KDD	1998	41	5	77054	71463
UNSW-NB15	2015	49	10	2218761	321283
BoT-IoT	2018	46	5	477	3668045
CICIDS 2017	2018	77	7	2273097	557646
CSE-CIC-IDS2018	2018	75	-	2856035	1669364
WUSTL-IIoT-2021	2021	-	5	1106747	87014
KDDCup-99	1998	41	23	-	-
Edge-IIoT	2022	-	15	1091198	9728708
WSN-DS	2016	23	5	340066	34595
WUSTL-EHMS-2020	2020	44	-	14272	2046
MQTT-IoT-IDS2020	2020	83	5	-	-
BoTNeTIoT-L01	2019	34	2	-	-
NF-BoT-IoT	2020	12	5	13859	586241
NF-ToN-IoT	2020	12	10	270279	1108995
NF-CSE-CIC-IDS2018	2020	12	7	7373198	1019203
NF-UNSW-NB15	2020	12	9	1550712	72406
NF-UQ-NIDS	2020	12	21	9208048	2786845
InSDN	2020	-	8	343939	275515
CICIDS 2023	2023	46	8	1098195	45588384
IoT Intrusion	2019	83	4	40073	5805710
DS2OS	2018	13	8	357962	10027
AWID	2015	154	4	2163975	207243
IoT-23	2020	21	10	-	-
CIC-DDoS2019	2019	88	2	50000	12000000
X-IIoTID	2021	68	-	421417	399417

As we can remake in [Table 1](#), more than 20 datasets are published between 2018 and 2023. Hence, these datasets comprise various features that vary between 12 and 88 features. Furthermore, the total number of included attacks exceeds thousands, ranging from 2046 to 45,588,384 attacks. It is also clear that some recent datasets, like NF-CSE-CIC-IDS2018 (2020) and NF-UQ-NIDS (2020), comprise millions of records. Such diversity provides the opportunity for researchers to develop and implement IDS solutions that can handle a vast range of different scenarios in IoT environments.

Adopted deep learning algorithms

In our days, the increasing number of IoT equipment has raised the importance of strong security measures, particularly for detecting intrusions. By the way, numerous different approaches and algorithms have been developed or tested to address the particular challenges of today's IoT systems, focusing on intrusion detection in a variety of datasets. The reviewed studies indicate a diverse range of methods, including CNN, LSTM, and even more complex hybrid models and combined systems. Such algorithms are employed on datasets including CSE-CIC-IDS2018, BoT-IoT, and TON-IOT, illustrating the variety and progress of approaches to improving intrusion detection in IoT systems. In this section, we review the main results of these studies and highlight the different approaches that are shaping the future of IoT security (Kilichev et al., 2024).

As depicted in [Table 2](#), various algorithms are used for implementing IDS in IoT, including CNN, LSTM, DNN, and their combinations. Hence, CNN is used in CSE-CIC-IDS2018, while hybrid models like CNN-LSTM are used in BoT-IoT and CICIDS 2017. Advanced algorithms like AE and RNN-GRU are also used. AE is used in CSE-CIC-IDS2018, and RNN-GRU is used in TON-IOT. Integrated models like Deep Integrated Stacking for IoT (DIS-IoT) combining MLP, DNN, CNN, and LSTM are used in TON-IOT and CICIDS 2017.

It is clear that the reviewed papers demonstrate variety of algorithms and methods, ranging from CNNs and LSTMs to more complex hybrid models. Such approaches were applied to various IoT datasets. Things that demonstrate that researchers are based on different strategies to improve IoT security. Hence, this reflects the continuous development of techniques to meet the increasing challenges of protecting IoT networks.

Performance analysis of the different models studied

As the IoT continues to grow, ensuring security in these environments has become increasingly challenging. Many different algorithms and techniques have been developed to detect and prevent intrusions. Each method has its own strengths and weaknesses, depending on the situation. This section looks at the various studies and approaches used to address this issue, particularly focusing on network and host architectures ([Table 2](#)). The studies cover different algorithms such as CNN, LSTM networks, and hybrid models, showing how a variety of techniques are being used to improve IoT security (Kilichev et al., 2024).

Table 2 Performance of the different models studied.

Paper	Algorithm	Dataset	Accuracy	Precision	Recall	F1-Score
Awajan (2023)	FC	Average of five attacks	93,74%	93,71%	93,82%	93,47%
P Maheswaravenkatesh (2024)	CNN	CSE-CIC-IDS2018	99,65%	99,16%	98,70%	99,09%
Jose & Jose (2023)	CNN-LSTM	CSE-CIC-IDS2018, MQTT-IoT-IDS2020, BotNetIoT-L01	99,86%	99,80%	99,87%	99,83%
Devendiran & Turukmane (2024)	Dugat-LSTM	TON-IOT, NSL-KDD	98,76%	96,98%	97,87%	97,23%
Hazman, Guezzaz, Benkirane, & Azrou (2024)	AE, MI, GA, LSTM	BoT-IoT	99,94%	99,94%	99,94%	-
		Edge-IIoT	99,99%	100,00%	100,00%	-
		NSL-KDD	99,71%	1,84%	99,71%	-
Wardana et al. (2024)	DNN	N-BaloT	97,21%	91,41%	87,31%	88,48%
Abbas et al. (2024)	CNN + DNN + RNN	CICDIoT2023 - CNN3	96,37%	96,15%	96,37%	95,51%
		CICDIoT2023 - DNN3	88,64%	91,20%	88,64%	88,51%
		CICDIoT2023 - RNN1	96,52%	96,25%	96,52%	95,73%
Sharma et al. (2024)	CNN + DNN	UNSW-NBnew	81,00%	-	-	-
		KDD-CUP	99,20%	-	-	-
		UNSW-NB	81,00%	-	-	-
Parimala & Kayalvizhi (2024)	ERNN	-	98.52	96,00%	98,00%	-
Nazir et al. (2024)	CNN-LSTM hybride	IoT-23	95,00%	-	-	-
		CICID2017	98,99%	-	-	-
		N-BaloT	99,99%	-	-	-
Zawaideh et al. (2024)	CNN	NF-bot-IoT		100,00%	80,00%	89,00%

Table2 (Continued).

Paper	Algorithm	Dataset	Accuracy	Precision	Recall	F1-Score
Dina et al. (2023)	FNN-Focal + CNN-Focal	Bot-IoT	91,55%	55,59%	63,80%	57,84%
			86,77%	61,65%	63,25%	58,53%
		WUSTL-EHMS-2020	93,26%	95,24%	73,69%	80,11%
			93,08%	94,23%	73,38%	79,63%
		WUSTL-IloT-2021	98,95%	77,22%	64,06%	68,48%
			98,21%	88,54%	66,51%	70,50%
Catillo et al. (2023)	AE	Traffic from different IoT + Bot-IoT devices	-	99,99%–100%	99,94%–99,97%	99,96%–99,98%
	RNN-GRU	ToN-IoT	99,00%	99,00%	98,00%	97,00%
Lazzarini et al. (2023)	DIS-IoT (MLP, DNN, CNN, LSTM)	Binary classification	99,60%	99,40%	99,40%	99,40%
		CICIDS2017	98,70%	95,90%	97,60%	96,70%
		SWaT	99,60%	99,70%	99,90%	99,80%
Keshk et al. (2023)	LSTM	Binary classification	81,10%	92,10%	73,20%	81,50%
		NSL-KDD	86,60%	81,10%	98,80%	89,10%
		UNSW-NB15	87,30%	78,40%	88,00%	82,90%
Bakhsh et al. (2023)	FFNN, LSTM, RandNN	ToN_IoT	99,84%	99,93%	99,93%	99,93%
		FFNN	99,78%	99,89%	99,89%	99,89%
		LSTM	96,42%	96,42%	96,42%	96,42%
Soliman et al. (2023)	SVD + (LSTM, Bagging tree, Bi-LSTM, KNN, GRU)	Binary classification	99,99%	-	-	-
		Multiclass classification	99,98%	-	-	-
Odeh & Abu Taleb (2023)	CNN-LSTM, CNN-GRU	CNN-LSTM	99,73%	99,70%	99,90%	99,80%
		CNN-GRU	99,60%	99,50%	99,90%	99,70%
Elmakib et al. (2023)	EIDM (MLP, CNN, LSTM, CNN + LSTM)	CICIDS2017	95,00%	-	-	-
		Multiclass classification				
Baniasadi et al. (2022)	NSBPSO-DCNN	-	98,86%	99,03%	-	-
Qazi et al. (2022)	AE	-	99,65%	99,99%	99,85%	99,55%

Table 2 (Continued).

Paper	Algorithm	Dataset	Inception time	Accuracy	Precision	Recall	F1-Score
Tareq et al. (2022)	DenseNet and inception time(CNN)	ToN-IoT Edge-IoT	DenseNet	97,7%-99,9%	95,97%-99,9%	95,91%-99,9%	95,91%-99,9%
Sharma et al. (2023)	GAN-DNN	UNSW-NB15 UNSW-NB15	DenseNet DNN	94,94% 98,60%	98,30% 98,90%	92,40% 98,40%	95,30% 98,70%
Saba et al. (2022)	GAN-DNN	UNSW-NB15	GAN-DNN	84,00%	-	-	-
		NID	GAN-DNN	91,00%	-	-	-
	CNN	Bot-IoT	NID	99,51%	-	-	-
Halbouni et al. (2022)	CNN-LSTM	CIC-IDS2017	Binary classification	95,55%	-	-	-
			Multiclass classification	99,64%	-	-	99,56%
			Multiclass classification	99,60%	-	-	99,60%
		UNSW-NB15	Binary classification	94,53%	-	-	94,69%
			Multiclass classification	82,41%	-	-	94,77%
		WSN-DS	Binary classification	99,67%	-	-	98,00%
			Multiclass classification	98,83%	-	-	98,44%
Banaamah & Ahmad (2022)	CNN,LSTM,GRU	Bot-IoT	CNN	99,70%	99,60%	99,90%	99,80%
			LSTM	99,80%	99,70%	100,00%	99,80%
Jose & Jose (2023)	DNN, LSTM, CNN	CIC-IDS2017	GRU	99,60%	99,60%	100,00%	99,80%
			DNN	94,61%	80,85%	84,60%	84,60%
			LSTM	97,67%	94,96%	95,00%	93,55%
Alimi et al. (2022)	RLSTM	NSL-KDD	CNN	98,61%	97,05%	96,95%	98,09%
			Average (DOS attack, Normal)	98,60%	98,60%	98,60%	98,60%
		CICIDS-2017	Average(DOS attack, Normal)	99,20%	99,23%	99,22%	99,22%
Vishwakarma & Kesswani (2022)	-	NetFlow(NF-BoT-IoT, NF-ToN-IoT, NF-CSE-CIC-IDS2018, NF-UNSW-NB15, NF-UQ-NIDS)		93,02%	-	-	-

Table2 (Continued).

Paper	Algorithm	Dataset	Accuracy	Precision	Recall	F1-Score
Ahmad et al. (2022)	TCN; AE - TCN; AE - LSTM; AE - BRNN; AE - BLSTM;CNN - LSTM	Bot-IoT	100%	-	-	-
		CICIDS2017	99.90%	-	-	-
		NSL-KDD	79.00%	-	-	-
		UNSW_NB15	97.90%	-	-	-
Balaji et al. (2024)	GAN	N-BaloT	90.90%	-	-	-
		KDD CUP 99	99.90%	-	-	-
		Aggregated Dataset	98%	98%	-	-
		ToN-IoT	97.5%	98.4%	97.9%	98.05%
Elsayed et al. (2022)	LSTM	InSDN	99%	-	99.60%	99.3%
		Vehicle network packets	91.26%	-	90.96%	-
R et al. (2024)	DBN	UNSW-NB15	83.18%	-	83.70%	81.19%
Kocher & Kumar (2022)	CNN-Bi-LSTM	ToN-IoT (Multiclass classification)	88%	86%	97%	88%
Khan et al. (2023a)	RNN-GRU	ToN-IoT (Binary classification)	99%	98%	99%	98%
Kumar & Sharma (2023)	IHLNA	KDD-CUP99	99.56%	99.67%	-	-
		NSLKDD	99.96%	99.67%	-	-
		UNSW-NB-15	99.96%	99.67%	-	-
		NSL-KDD	91.70%	81.50%	79.88%	80.68%
Yang et al. (2024)	DPA-LSCNN	CIC IDS2017	99.67%	99.54%	99.67%	99.59%
Wang et al. (2023)	DNN-BiLSTM	N-BaloT	99.98%	99.98%	99.98%	99.98%
		CIC IoT2023	93.13%	91.80%	93.13%	91.94%
Idrissi et al. (2022)	CNN	MQTTIoT-IDS2020	99.74%	-	-	-
		UNSW-NB-15 (AAFSa with GA-FR-CNN)	94.48%	94.29%	94.56%	94.42%
Anushiya & Lavanya (2023)	FR-CNN; GA-FR-CNN	BoT NeT IoT (AAFSa with GA-FR-CNN)	93.77%	86.66%	95.87%	91.03%
Hizal et al. (2024)	DNN-CNN-LSTM-RNN	CICIoT2023 (Binary)	99.76%	-	-	-
		CICIoT2023 (Multiclass)	91.27%	-	-	-
Maheswari & Karthika (2022)	SHO-LSTM	Benchmark datasets	99.89%	98%	97.5%	99%

Table 2 (Continued).

Paper	Algorithm	Dataset	Accuracy	Precision	Recall	F1-Score
Kim et al. (2024)	DTL	IoT Intrusion (Target: Bot-IoT)	99.94%	99.94%	100%	99.97%
Raviprasad et al. (2022)	CNN	Bot-IoT (Target: IoT Intrusion)	99.94%	99.94%	100%	99.97%
Rashid et al. (2022)	DNN	NSL-KDD	97.29%	-	-	-
	DT	DS2OS(multi-class)	99.41%	99%	99%	99%
	RF		99.44%	99%	99%	99%
Mahjoub et al. (2024)	AE-RL	BOT-IOT (Class:Normal)	99.44%	99%	99%	99%
		BOT-IOT (Class:DOS)	99.98%	42.46%	59.61%	100%
Salman et al. (2022)	CNN-LSTM	UNSW_NB15	94.59%	96.43%	94.76%	93.14%
	ANN		96.08%	96.08%	96.08%	96.08%
Cao et al. (2022)	CNN-SAE	AWID (Two classes)	97.01%	97.01%	97.01%	97.01%
Mohamed et al. (2024)	LSTM-RNN; DNN; ResNet	N-BalIoT	99.77%	97.95%	99.09%	98.51%
		UNSW-NB15	99.8%	-	-	-
X. Yang et al. (2022)	CNN-BiLSTM-Attention	NSL-KDD	99.82%	-	-	-
Alghamdi & Bellaiche (2023)	ANN-CNN-LSTM	IoT-23 (.DDOS)	90.01%	90.35%	91.07%	90.71%
			99.6%	93.5%	99.1%	96.3%
Alsoufi et al. (2024)	SAE-CNN	Bot-IoT	99.9%	99.9%	100%	99.9%
Rodríguez et al. (2023)	TabNet	CIC-IDS2017	97%	-	-	-
		CSE-CICIDS2018	95%	-	-	-
		CIC-DDoS2019	98%	-	-	-
Henry et al. (2023)	CNN-GRU	CIC-IDS2017	98.73%	-	-	-
Chaganti et al. (2023)	LSTM	SDN-IoT, SDN-NF-TJ	97.1%	-	-	-
Otoom et al. (2023)	Algorithm not mentioned	MQTT-IoT-IDS2020 (Bi-flow)	99.56%	99.60%	99.60%	99.60%
		MQTT-IoT-IDS2020 (Uni-flow)	99.67%	99.70%	99.70%	99.70%
Shah et al. (2023)	RF-LSTM-GRU-ANN	X-IloTID	99.72%			

Table2 (Continued).

Paper	Algorithm	Dataset	Accuracy	Precision	Recall	F1-Score
Hazman et al. (2025)	LSTM-PCC-XGBoost	Bot-IoT (LSTM) (binary classification) EdgeloT (PCC-LSTM) (binary classification)	99.99% 99.99%	99.99% 99.99%	100% 100%	99.99% 98.26%
Bella et al. (2024)	DNDF (CNN variant)	NSL-KDD (PCA) CICIDS2017 (PCA) UNSW-NB15 (PCA)	98.38 98.84% 98.23%	98.08% 99.12% 96.90%	98.69% 98.56% 99.65%	98.38% 98.84% 98.25%
C. Hazman, A. Guezzaz, S. Benkirane, & M. Azrour (2024)	LSTM-TPU	BoT-IoT Edge-IIoT NSL-KDD	99.94% 99.99% 99.71%	99.94% 100% 01.84%	99.94% 100% 99.71%	- - -
Nobakht et al. (2023) Sanju (2023)	CNN LSTM-GRU	IoT-23 (Meta-learner: RF) IoT23 UNSW-NB15	99.90% 98.12% 99.98%	99.83% 98.06% 99.99%	99.97% 98.31% 99.98%	99.90% 98.18% 99.99%

Many studies have shown the optimal algorithms and datasets for intrusion detection tasks. For binary classification, LSTM-PCC-XGBoost achieved 99.99% accuracy on the Bot-IoT dataset, as mentioned in (Hazman et al., 2025). Similarly, IHLNA achieved impressive results on the NSL-KDD dataset, with a 99.96% accuracy as reported in (Kumar & Sharma, 2023). For the MQTT-IoT-IDS2020 dataset (Uni-flow), an unknown algorithm got 99.70% accuracy as described by (Otoom et al., 2023). These findings demonstrate the capability of some of the models to address the binary classification problems in various datasets.

For multiclass classification problems, in (Alghamdi & Bellaiche, 2023) CNN-LSTM has been shown to be effective with a 99.60% accuracy on the CIC-IDS2017 data as reported there. On the UNSW-NB15 dataset, CNN-LSTM also performed well, achieving 94.77% accuracy, as reported in (Salman et al., 2022). The DNN-BiLSTM model worked very well on the CICIOT2023 dataset with 93.13% classification accuracy, as reported in (Wang et al., 2023). These case studies show that models using CNN and LSTM components are appropriate for the multiclass classification task in network intrusion detection.

Among the most popular datasets, NSL-KDD has been widely evaluated. IHLNA reached 99.96% accuracy on this dataset, as described in (Kumar & Sharma, 2023). The model of CNN-BiLSTM-Attention achieved 91.07% accuracy as reported by (Yang et al., 2022). The DNDF model, combined with PCA preprocessing, obtained an accuracy of 98.38%, as highlighted in (Bella et al., 2024). Among the CIC-IDS2017, the DNN-BiLSTM obtained the highest (99.67% score (Z. Wang et al., 2023), as well as the second-highest (98.73% score (Henry et al., 2023), respectively. UNSW-NB15 has also demonstrated potential application to LSTM-GRU achieving 99.99% as described in (Sanju, 2023). And these datasets still work as benchmarks for evaluating and optimizing the models in the community.

In general, hybrids architectures such as CNN-LSTM, DNN-BiLSTM, and LSTM-PCC-XGBoost are superior to pure models for addressing difficult intrusion detection problems. Papers e.g (Alghamdi & Bellaiche, 2023; Hazman et al., 2025; Wang et al., 2023) demonstrate the benefits of use of these integrative methods. Further, by way of example, PCA-based pre-processing (as described in (Bella et al., 2024) and the inclusion of datasets such as Bot-IoT and MQTT-IoT-IDS2020, which are related to the Internet of Things, show the direction of the field—toward solving IoT security problems. These studies also highlight the requirement of new models for effectively countering emerging threats.

This research compilation offers an overview of the latest advances in the area of intrusion detection, showcasing the different approaches and algorithms employed by researchers to enhance security in IoT environments, where security challenges are particularly complex and evolving.

Analysis of intrusion detection systems: Architectures, algorithms, and methodologies

Studies reporting on IDS and architectures and algorithms based on them are reported in Table 3. All rows represent an IDS

Table 3 Detailed summary of algorithms, methodologies, advantages, and limitations from reviewed papers.

Paper	Model Characteristics
(Awajan (2023))	Algorithm used: FC. Dataset used: Model trained with the dataset produced by the experimental system. Language Used: Python. Methodology: Developed a four-layer deep fully connected neural network; Attack detection; Experimental validation. Advantages: High accuracy; Protocol independence; Broad attack coverage; Real-time detection. Disadvantages: Dataset dependence; Computational complexity; Limited attack types; Potential false positives.
Murugesan & Maheswaravenkatesh (2024)	Algorithm used: CNN. Dataset used: CSE-CIC-IDS2018. Language Used: Python. Methodology: Dataset preprocessing; Model architecture (Five convolutional layers); Experimental validation. Advantages: High performance; Automated feature extraction; Wide applicability. Disadvantages: Dataset dependency; Potential overfitting; Limited interpretability.
Jose & Jose (2023)	Algorithm used: CNN-LSTM. Dataset used: CIC-IDS2018, MQTT-IoT-IDS2020, BoTNeT-IoT-L01. Language Used: Python. Methodology: Data preprocessing; Model training and testing; Assessment. Advantages: High accuracy; Dual detection strategies; Feature reduction; Generalization. Disadvantages: Computational cost; Dataset dependency; Limited real-time validation; Scalability concerns.
Devendiran & Turukmane (2024)	Algorithm used: Dugat-LSTM. Dataset used: TON-IOT, NSL-KDD. Language Used: Not mentioned. Methodology: Preprocessing; Balancing; Feature handling; Model training; Evaluation. Advantages: High accuracy; Class imbalance handling; Feature optimization; Robust model. Disadvantages: Complexity; Computational cost; Generalization challenges; Dependence on preprocessing.
Hazman, Guezzaz, Benkirane, & Azroul (2024)	Algorithm used: AE; MI; GA; LSTM. Dataset used: BoT-IoT; Edge-IIoT; NSL-KDD. Language Used: Python. Methodology: Dataset preparation; Feature engineering; Model architecture; Implementation; Evaluation. Advantages: High accuracy; Low latency; Adaptability; Temporal analysis. Disadvantages: Resource dependency; Complexity; Scalability challenges; Dataset-specific performance.
Wardana et al. (2024)	Algorithm used: DNN. Dataset used: N-BalIoT. Language Used: Python. Methodology: Dataset preparation; Model training; Ensemble averaging; Validation. Advantages: High accuracy; Adaptability; Improved generalization; Scalability. Disadvantages: Computational overhead; Latency; Dataset dependency; Complexity.

Table3 (Continued).

Paper	Model Characteristics
Abbas et al. (2024)	Algorithm used: CNN + DNN + RNN. Dataset used: CICDIoT2023. Language Used: Not mentioned. Methodology: Dataset preparation; Model Implementation; Evaluation. Advantages: High detection accuracy; Temporal pattern recognition; Applicability to realistic scenarios. Disadvantages: Computational intensity; Potential overfitting; Limited generalization.
Sharma et al. (2024)	Algorithm used: CNN + DNN. Dataset used: UNSW-NBnew; KDD-CUP; UNSW-NB. Language Used: Not mentioned. Methodology: Dataset Preparation; Model development; Explainability. Advantages: High classification accuracy; Feature reduction; Explainability. Disadvantages: Computational complexity; Generalization; Interpretability limitations.
Parimala & Kayalvizhi (2024)	Algorithm used: ERNN. Dataset used: KDDCup-99; NSL-KDD. Language Used: MATLAB. Methodology: Dataset preparation; Feature selection; Model development; Performance evaluation. Advantages: High accuracy; Effective feature selection; Robust optimization. Disadvantages: Dataset limitations; Computational complexity; Generalization.
Nazir et al. (2024)	Algorithm used: Hybrid CNN-LSTM. Dataset used: IoT-23; CICID2017; N-BalIoT. Language Used: Not mentioned. Methodology: Dataset preparation; Feature extraction and modeling; Model optimization; Validation and testing. Advantages: High accuracy; Adaptability; Efficient deployment. Disadvantages: Model complexity; Limited real-world validation; Dependence on PCA.
Zawaideh et al.	Algorithm used: CNN. Dataset used: NF-bot-IoT. Language Used: Python. Methodology: Data collection; Feature extraction with CNN; Classification with XGBoost; Model training and testing. Advantages: High detection accuracy; Effective feature extraction; Scalability. Disadvantages: Computational power; Data dependency; Model complexity.
Dina et al. (2023)	Algorithm used: FNN-Focal + CNN-Focal. Dataset used: Bot-IoT; WUSTL-EHMS-2020; WUSTL-IIoT-2021. Language Used: Not mentioned. Methodology: Data collection and preprocessing; Handling data Imbalance with focal loss; Training and evaluation; Comparison with state-of-the-art approaches. Advantages: Improved performance on imbalanced data; Better generalization; State-of-the-art comparison. Disadvantages: Computational complexity; Dataset dependency; Model interpretability.

Table3 (Continued).

Paper	Model Characteristics
Catillo et al. (2023)	Algorithm used: AE Dataset used: Traffic from different IoT + Bot-IoT devices Language Used: Python. Methodology: Data preparation; Model architecture; Detection; Evaluation. Advantages: Device independence; Lightweight; Transferability; High accuracy. Disadvantages: Semi-supervised limitation; Unseen anomalies; False positive sensitivity.
Khan (2023)	Algorithm used: RNN-GRU. Dataset used: ToN-IoT. Language Used: Python. Methodology: Three-layered IoT system; The model consists of RNN-GRU networks; Training and Testing; Optimization; Comparison with Other Techniques. Advantages: Cross-Layer Attack detection; High accuracy; Adaptability to new attacks; Optimization efficiency. Disadvantages: Resource intensity; Sensitivity to dataset quality; Complexity of hyperparameter tuning.
Lazzarini et al. (2023)	Algorithm used: MLP, DNN, CNN, and LSTM. Dataset used: ToN_IoT; CICIDS2017; SWaT. Language Used: Python. Methodology: Stacking ensemble approach; Model training; Comparison with other models. Advantages: Improved performance with ensemble learning; Effective multi-class classification; Low false positive rate; Scalability. Disadvantages: Complexity; Resource consumption; Dependence on high-quality data.
Keshk et al. (2023)	Algorithm used: LSTM. Dataset used: NSL-KDD, UNSW-NB15, ToN_IoT. Language Used: Python. Methodology: Model training; Explainability via SPIP; Evaluation. Advantages: High detection accuracy; Interpretability; Real-time capability; Generalization. Disadvantages: Complexity of model; Dependence on quality data; Interpretability overhead.
Bakhsh et al. (2023)	Algorithm used: FFNN, LSTM, RandNN. Dataset used: CIC-IoT22. Language Used: Not mentioned. Methodology: Model development; Training and testing; Comparison. Advantages: High accuracy; Adaptability; Fast response time; Wide applicability. Disadvantages: Computational complexity; Overfitting risk; Model complexity.
Soliman et al. (2023)	Algorithm used: SVD + (LSTM, Bagging tree, Bi-LSTM, KNN, GRU). Dataset used: TON-IOT. Language Used: Not mentioned. Methodology: Data preprocessing; Model training; Evaluation. Advantages: High accuracy; Addressing class Imbalance; Feature reduction for efficiency; Versatility for binary and multi-class classification. Disadvantages: Dependency on the ToN_IoT dataset; Computational complexity; Overfitting risk; Scalability in large-scale deployments.

Table3 (Continued).

Paper	Model Characteristics
Odeh & Abu Taleb (2023)	Algorithm used: CNN-LSTM, CNN-GRU. Dataset used: NSL-KDD. Language Used: Not mentioned. Methodology: Data preprocessing; Model training; Model evaluation. Advantages: High performance; Effective at capturing sequential patterns; Hierarchical feature learning; Robustness to variability. Disadvantages: High computational cost; Complexity; Data requirements; Risk of overfitting; Interpretability.
Elnakib et al. (2023)	Algorithm used: EIDM (MLP, CNN, LSTM, CNN + LSTM). Dataset used: CICIDS 2017. Language Used: Python. Methodology: Data preprocessing; Model training; Model evaluation. Advantages: High accuracy; Multi-class classification; Deep learning approach; Comparison with other models. Disadvantages: Computational complexity; Data dependency; Overfitting risk; Interpretability; Time complexity for real-time detection.
Baniasadi et al. (2022)	Algorithm used: NSBPSO-DCNN. Dataset used: BoT-IoT, UNSW-NB15. Language Used: MATLAB. Methodology: Algorithm design; Data preprocessing; Model training; Model evaluation. Advantages: Improved optimization; Higher detection accuracy; Adaptability; Hybrid optimization. Disadvantages: Computational complexity; Dependence on data quality; Overfitting risk; Real-time processing.
Qazi et al. (2022)	Algorithm used: AE. Dataset used: KDDCup-99. Language Used: Python. Methodology: Model design; Training process; Validation; Implementation. Advantages: Offers high accuracy, precision, recall, and F1-score with reduced training time Disadvantages: Computational complexity; The results obtained are less stable due to the reduced number of training data.
Tareq et al. (2022)	Algorithm used: DenseNet and Inception Time (CNN). Dataset used: ToN-IoT, Edge-IIoT, UNSW-NB15. Language Used: Not mentioned. Methodology: Data preprocessing; Model training; Evaluation and comparison; Sliding window approach (Inception Time). Advantages: High accuracy; Versatility; Time-series data handling; Effective multi-class classification; Disadvantages: Computational cost; Overfitting risk; Data quality and representation; Interpretability.
Sharma et al. (2023)	Algorithm used: GAN-DNN. Dataset used: UNSW-NB15. Language Used: Python. Methodology: Data preprocessing; Model training; Class imbalance solution; Evaluation. Advantages: High accuracy after balancing; Improved performance with feature selection; Handling class imbalance. Disadvantages: Computational cost; Overfitting risk with GANs; Data quality and representativeness; Interpretability of the DNN model.

Table3 (Continued).

Paper	Model Characteristics
Saba et al. (2022)	Algorithm used: CNN. Dataset used: Bot-IoT. Language Used: Python. Methodology: Data preprocessing; Model architecture; Training and testing. Advantages: High accuracy; Scalability; Adaptability to IoT traffic; Ability to detect complex attacks. Disadvantages: Computational complexity; Overfitting risk; Interpretability; Data dependency.
Halbouni et al. (2022)	Algorithm used: CNN-LSTM. Dataset used: CIC-IDS2017, UNSW-NB15, WSN-DS. Language Used: Python. Methodology: Data preprocessing; Model architecture; Model training; Model evaluation. Advantages: High detection rate; Robust performance; Automated feature extraction; Reduced false alarm rate. Disadvantages: Computationally expensive; Data dependency; Complexity of model tuning; Risk of overfitting.
Banaamah & Ahmad (2022)	Algorithm used: CNN, LSTM, GRU. Dataset used: Bot-IoT. Language Used: Python. Methodology: Data preprocessing; Model training; Model evaluation. Advantages: High accuracy; Deep learning advantages; Ability to handle complex patterns; Scalability. Disadvantages: Computational complexity; Data dependency; Overfitting risk; Interpretability.
Jose & Jose (2023)	Algorithm used: DNN, LSTM, CNN. Dataset used: CIC-IDS2017. Language Used: Not mentioned. Methodology: Model development; Comparison. Advantages: High accuracy; Scalability; Flexibility; Feature learning. Disadvantages: Computational cost; Overfitting risk; Data dependency; Interpretability.
Alimi et al. (2022)	Algorithm used: RLSTM. Dataset used: NSL-KDD. Language Used: MATLAB. Methodology: Data preprocessing; Model training; Performance comparison. Advantages: High performance; Temporal dependency learning; Reduced human intervention; Scalability. Disadvantages: Computational cost; Data dependency; Overfitting risk; Interpretability.
Vishwakarma & Kesswani (2022)	Algorithm used: DNN. Dataset used: NF-UQ-NIDS, NF-UNSW-NB15, NF-CSE-CIC-IDS2018. Language Used: Python. Methodology: Packet capturing and detection; Dataset preparation; Model training and evaluation. Advantages: Real-time detection; High accuracy; Suitability for IoT constraints; Automatic feature extraction. Disadvantages: High computational requirements; Potential for overfitting; Limited interpretability; Dependency on dataset quality.

Table3 (Continued).

Paper	Model Characteristics
Ahmad et al. (2022)	Algorithm used: TCN;Autoencoder - TCN;Autoencoder - LSTM;Autoencoder - BRNN;Autoencoder - BLSTM;CNN – LSTM. Dataset used: Bot-IoT; CICIDS2017; NSL-KDD; UNSW_NB15; N-BaloT; KDD CUP 99. Language Used: Not mentioned. Methodology: Dataset analysis; Classifier evaluation; Comparison and benchmarking; Empirical results. Advantages: Comprehensive evaluation; Reduced bias; Improved IDS design. Disadvantages: Computationally intensive; Dependency on dataset quality; Complexity.
Balaji et al. (2024)	Algorithm used: GAN. Dataset used: Aggregated Dataset. Language Used: Python. Methodology: Problem addressing; Model development; Preprocessing; Evaluation. Advantages: Improved detection; Reduced false positives; Scalability; Data augmentation. Disadvantages: Computational complexity; Threshold sensitivity; Model interpretability; IoT-specific scope.
Elsayed et al. (2022)	Algorithm used: LSTM. Dataset used: ToN-IoT; InSDN. Language Used: MATLAB. Methodology: Feature selection; Model development; Performance analysis; Routing protocol integration. Advantages: High performance; Feature selection; IoT-specific focus; Integration with routing protocols. Disadvantages: Computational requirements; Complexity; Dataset dependency; Scalability.
Priyanka et al. (2024)	Algorithm used: DNN-DBN. Dataset used: Vehicle network packets. Language Used: MATLAB. Methodology: Model training; Integration with DBN; Decision reporting; Real-time evaluation. Advantages: High detection rate; Hierarchical clustering; Real-time capabilities; Comprehensive analysis; Decision-making support. Disadvantages: Computational requirements; False positives; Dataset dependency; Limited generalization; Cluster head vulnerability.
Kocher & Kumar (2022)	Algorithm used: CNN-Bi-LSTM. Dataset used: UNSW-NB15. Language Used: Python. Methodology: Data preprocessing; Model design; Model training and testing; Comparative analysis. Advantages: Higher accuracy; Improved precision; Low false positive rate; Comprehensive feature Learning; State-of-the-art performance. Disadvantages: Complexity; Training time; Data dependency; Overfitting risk; Interpretability.
Khan et al. (2023a)	Algorithm used: RNN-GRU. Dataset used: ToN-IoT. Language Used: Python. Methodology: Data collection; Model design; Training and testing; Comparison with Other techniques. Advantages: Enhanced performance; Improved data processing; Real-world applicability. Disadvantages: Dataset dependency; High computational requirements; Limited real-world testing.

Table3 (Continued).

Paper	Model Characteristics
Kumar & Sharma (2023)	Algorithm used: IHLNA. Dataset used: KDD-CUP99; NSLKDD; UNSW-NB-15. Language Used: MATLAB. Methodology: Data preprocessing and feature selection; Data up-sampling; Layered network model; Model training. Advantages: High accuracy and performance; Low false acceptance rate; Cost-effective. Disadvantages: Dataset dependency; Limited real-world validation; Potential overfitting.
Yang et al. (2024)	Algorithm used: DPA-LSCNN. Dataset used: NSL-KDD. Language Used: Not mentioned. Methodology: Data purification (DPA); Conversion of data to image data; Separable convolutions; LSCNN model. Advantages: Improved accuracy; Reduced computational cost; Enhanced feature extraction. Disadvantages: Increased model complexity; Training time limitations; Attribute correlation issue.
Wang et al. (2023)	Algorithm used: DNN-BiLSTM. Dataset used: CIC IDS2017; N-BaIoT; CICIOT2023. Language Used: Python. Methodology: Model architecture; Feature dimensionality reduction; Dynamic quantization; Training and evaluation. Advantages: Enhanced detection accuracy. Disadvantages: Implementation complexity; Dataset specificity.
Idrissi et al. (2022)	Algorithm used: CNN. Dataset used: MQTTIOT-IDS2020. Language Used: Python. Methodology: Device-specific optimizations; Model deployment; Centralized IDS in fog or cloud layers; Evaluation on multiple devices. Advantages: High accuracy; Real-time Detection; Lightweight optimization. Disadvantages: Device dependency; Limited dataset information.
Anushiya & Lavanya (2023)	Algorithm used: FR-CNN; GA-FR-CNN. Dataset used: UNSW-NB 15; BoT NeT IoT. Language Used: MATLAB. Methodology: Feature selection with AAFSO; Model training with GA-FR-CNN; Dataset evaluation. Advantages: High accuracy; Optimized feature selection; Generalizability. Disadvantages: Complexity of GA-FR-CNN; Dataset dependency; Training time.
Hizal et al. (2024)	Algorithm used: DNN-CNN-LSTM-RNN. Dataset used: CICIOT2023. Language Used: Python. Methodology: Preprocessing operations; Model training. Advantages: Improved detection efficiency; Focused attack identification. Disadvantages: Moderate accuracy; Scalability concerns; Vulnerability to adversarial attacks.
Maheswari & Karthika (2022)	Algorithm used: SHO-LSTM. Dataset used: Aggregated Dataset. Language Used: Python. Methodology: Hybrid deep learning model; Real-time experimentation; Preprocessing and optimization; Testing on multiple datasets. Advantages: High detection accuracy; Comprehensive performance; Scalability. Disadvantages: Performance drop in real-time scenarios; Energy consumption considerations.

Table3 (Continued).

Paper	Model Characteristics
Kim et al. (2024)	Algorithm used: DTL. Dataset used: IoT Intrusion; Bot-IoT. Language Used: Python. Methodology: Transfer learning framework; Dataset selection process; Pre-training using transfer learning; Implementation and evaluation. Advantages: Efficient selection of source domain; Universal applicability; Extensibility. Disadvantages: Dependence on dataset selection; Time-to-accuracy optimization; Lack of detail on neural network architecture.
Raviprasad et al. (2022)	Algorithm used: DCN. Dataset used: NSL-KDD. Language Used: Not mentioned. Methodology: Deep convolution network (DCN) IDS; Deep learning for IDS; Multicloud IoT integration; Optimization of training. Advantages: Better overall performance; Multi-level feature extraction; Reduced training time. Disadvantages: Computational complexity; Dataset dependence; Limited generalization.
Rashid et al. (2022)	Algorithm used: DNN-DT-RF. Dataset used: DS2OS. Language Used: Python. Methodology: Model training; Adversarial sample generation and retraining. Advantages: Enhanced resilience; Generalization; Practical application. Disadvantages: Sensitivity to adversarial attacks; Complexity; Feature engineering challenges.
Mahjoub et al. (2024)	Algorithm used: AE-RL. Dataset used: BOT-IOT. Language Used: Not mentioned. Methodology: Simultaneous fine-tuning of the environment; Classifier embedded in the RL model; Evaluation using Bot-IoT dataset. Advantages: Improved performance; Adversarial strategy; Innovative framework. Disadvantages: Computational complexity; Dataset dependency; Potential overfitting.
Salman et al. (2022)	Algorithm used: CNN-LSTM; ANN. Dataset used: UNSW_NB15. Language Used: Python. Methodology: Model 1: CNN-LSTM; Model 2: ANN with fully connected layers; Training and evaluation. Advantages: High performance; Hybrid approach; Custom architecture. Disadvantages: Dataset dependency; Computational cost; Model complexity.
Cao et al. (2022)	Algorithm used: CNN-SAE. Dataset used: AWID. Language Used: Not mentioned. Methodology: Efficient data processing; Model design; Training and evaluation. Advantages: High accuracy; Lightweight architecture; Fast response time. Disadvantages: Dataset dependency; Limited attack types; Lack of robustness testing.
Mohamed et al. (2024)	Algorithm used: LSTM-RNN; DNN; ResNet. Dataset used: N-BalIoT; UNSW-NB15. Language Used: Not mentioned. Methodology: Data preprocessing; Ensemble strategy; Hyperparameter tuning; Comparison with other models. Advantages: Improved performance; Adaptability; Diversity of classifiers; Potential for zero-day attack detection. Disadvantages: Complexity; High computational resources; Risk of overfitting; Interpretability.

Table3 (Continued).

Paper	Model Characteristics
Yang et al. (2022)	Algorithm used: CNN-BiLSTM-Attention. Dataset used: NSL-KDD. Language Used: Python. Methodology: Feature extraction; Feature fusion and alignment; Model design; Evaluation. Advantages: High accuracy; Effective detection; Model interpretability. Disadvantages: High computational complexity and resource requirements; dataset dependency.
Alghamdi & Bellaiche (2023)	Algorithm used: ANN-CNN-LSTM. Dataset used: IoT-23. Language Used: Python. Methodology: Data preprocessing; Model design and training; Real-time analysis; Evaluation and comparison. Advantages: Scalability; Real-time detection; Flexibility. Disadvantages: Complexity; Resource intensive; Dataset dependency; Lack of real-world validation.
Alsoufi et al. (2024)	Algorithm used: SAE-CNN. Dataset used: Bot-IoT. Language Used: Python. Methodology: Data preprocessing; Feature extraction; Model design and training; Training and testing; Performance comparison. Advantages: High accuracy; Dimensionality reduction; Real-time capability; Low false positives. Disadvantages: Limited dataset scope; Binary classification; Resource intensive; Scalability concerns.
Rodríguez et al. (2023)	Algorithm used: TabNet. Dataset used: CIC-IDS2017; CSE-CICIDS2018; CIC-DDoS2019. Language Used: Python. Methodology: Data preprocessing; Feature selection; Model training; Evaluation. Advantages: High accuracy; Interpretability; Tabular data efficiency. Disadvantages: Resource intensive; Dataset dependency; Limited real-world validation; Complexity.
Henry et al. (2023)	Algorithm used: CNN-GRU. Dataset used: CIC-IDS2017. Language Used: Python. Methodology: Dataset preprocessing; Feature learning; Model training; Comparative analysis. Advantages: High accuracy; Feature learning; Reduced false alarms; Versatility. Disadvantages: Dataset dependency; Computational overhead; Explainability; Imbalanced dataset.
Chaganti et al. (2023)	Algorithm used: LSTM. Dataset used: SDN-IoT; SDN-NF-TJ. Language Used: Python. Methodology: Dataset preprocessing; Model design; Validation; Performance analysis. Advantages: Outperforms traditional ML and other DL models in classifying attacks; High generalizability. Disadvantages: vulnerable to adversarial attacks; Higher computational resource.
Otoom et al. (2023)	Algorithm used: Algorithm not mentioned. Dataset used: MQTT-IoT-IDS2020. Language Used: WEKA. Methodology: Data collection; Classification; Performance evaluation. Advantages: High detection accuracy; Focus on MQTT protocol; Scalability. Disadvantages: Binary classification limitation; Dataset-specific optimization; Interpretability.

Table3 (Continued).

Paper	Model Characteristics
Shah et al. (2023)	Algorithm used: RF-ANN-LSTM-GRU. Dataset used: X-IloTID. Language Used: Python. Methodology: Intrusion detection; Data security via blockchain; Smart contract classification. Advantages: High accuracy; Integration of AI and blockchain; Comprehensive security layers; Prevention mechanism. Disadvantages: Computational complexity; Scalability challenges; Adversarial vulnerability.
Hazman et al. (2025)	Algorithm used: LSTM-PCC-XGBoost. Dataset used: BoT-IoT; Edge-IloT. Language Used: Python. Methodology: Feature selection; Outlier detection; Classification; Evaluation; Optimization. Advantages: Improved performance on imbalanced datasets; Robust metrics; Scalability potential; Future enhancements. Disadvantages: Computational complexity; Edge computing optimization needed; Scalability challenges.
Bella et al. (2024)	Algorithm used: DNDF. Dataset used: NSL-KDD; CICIDS2017; UNSW-NB15. Language Used: Python. Methodology: Feature selection; Model development; Performance optimization; Comparative analysis. Advantages: High accuracy; Efficiency with limited features; Fast prediction; Versatility; Feature selection integration. Disadvantages: Dependency on feature selection; Complexity; Dataset variability; Scalability challenges.
Hazman, Guezzaz, Benkirane, & Azrour (2024)	Algorithm used: LSTM-TPU. Dataset used: Enhanced BoT-IoT; Edge-IloT; NSL-KDD. Language Used: Python. Methodology: Model development; TPU utilization; Dataset testing; Performance comparison. Advantages: High accuracy; Fast processing; Robust evaluation; Integration with IoT standards; Scalability. Disadvantages: Dependence on TPUs; Feature engineering complexity; Limited metric explanation; Dataset bias; Real-world challenges.
Nobakht et al. (2023)	Algorithm used: CNN. Dataset used: IoT-23. Language Used: Python. Methodology: Three one-dimensional convolutional neural networks are employed; Ensemble learning; Hyperparameter optimization. Advantages: High accuracy; Ensemble learning; Reduced processing time; IoT-specific design. Disadvantages: Model complexity; Dataset dependency; Training time.
Sanju (2023)	Algorithm used: LSTM-GRU. Dataset used: IoT-23; CICIDS2017. Language Used: Python. Methodology: Hybrid metaheuristics-deep learning approach; Feature selection; RNN models. Advantages: Improved intrusion detection; Optimized feature selection; Handling diverse IoT attacks. Disadvantages: Complexity; Dependence on feature selection; Scalability.

architecture implemented by a different study, including the algorithms, method of implementation, and the pros and cons of each approach.

IDS architectures and algorithms used: In this research, various IDS architectures, including that of network and host, as well as various algorithms including CNN, LSTM, autoencoders (AE), and deep neural networks (DNN) etc., are proposed.

Methodology: All the studies propose a different way for intrusion detection and usually are a mixture of models, even deep learning architectures.

Advantages: The benefit of the approaches varies, depending on the method, but some works report encouraging results, with respect to the accuracy, successful attack detection, real-time response and low computational complexity.

Disadvantages: The drawbacks listed are generally the complexity increases, attack types not detectable, model transfer limitations, and dataset.

Algorithms examined in the papers employ different deep learning and hybrid methods to perform intrusion detection in IoT systems. For example, the GA-FR-CNN and DNN-CNN-LSTM-RNN models integrate convolutional neural networks (CNNs) with long short-term memory (LSTM) networks, with good sensitivity and robustness in detection. Yet, such models, e.g., GA-FR-CNN in (Anushiya & Lavanya, 2023), are computationally expensive and demand heavy resources for training and inference. In particular, the effectiveness of such methods greatly depends upon the set of data used, meaning they are far from being easily generalizable to real-world deployments.

Computational efficiency is one of the most concern of its studies. Although deep learning models such as LSTM-RNN and RF-ANN-LSTM-GRU have been shown to deliver high accuracies, they require extensive computing time and computing power, which may be unsuitable in environments with limited computational power, as reported in the LSTM-RNN study (Chaganti et al., 2023). In contrast, models like TabNet in (Rodríguez et al., 2023) aim to reduce the computational load while maintaining strong performance. Yet even these more efficient models have scalability limitations, when used to work on bigger or more complex datasets. In addition, the possibility of adversarial attack on these models, as evidenced in work such as DNN-DT-RF (Kim et al., 2024), is another key issue to be addressed for deploying these models in security-critical IoT platforms.

Dataset dependency and the need for extensive preprocessing are recurring issues across the reviewed models. Some models, e.g., DTL (Kim et al., 2024) and CNN-LSTM (Anushiya & Lavanya, 2023), achieve promising performance on limited datasets, but the performance drops whenever they are applied to another/unseen data. This illustrates the role that borrowing eclectic data sets can play in achieving generalization of intrusion detection models. On the other hand, for certain models such as AE-RL (Mahjoub et al., 2024), complex feature engineering/preprocessing is necessary, so as to increase the system complexity, which may also restrict its practical use as a real-time system in dynamic and fast-changing IoT environment.

Interpretability and transparency are often sacrificed for improved accuracy in many models. In particular, although models such as CNN-BiLSTM-Attention (Yang et al., 2022) and LSTM-TPU (Hazman, Guezaz, Benkirane, & Azrou, 2024), while achieving high detection accuracy, are not transparent enough so as to be able to explain how the classifiers reason. This is especially critical in real-world applications, where the model's decision-making can be understood. Hybrid models, such as CNN-GRU in (Henry et al., 2023) and RF-ANN-LSTM-GRU (Shah et al., 2023), offer a balance between accuracy and robustness, but their complexity can pose challenges in terms of resource utilization and explainability, making them less suitable for environments with strict resource constraints.

These models are highly accurate and robust, but they often require significant hardware resources and continuous evaluation of their performance in real time.

Computational and resource constraints

In these various studies on intrusion detection in IoT using deep learning, the limits of computational capabilities are critical. IoT devices have restricted computing and storage resources, along with bandwidth constraints. To adapt, deep learning models must be tailored by simplifying algorithms and reducing complexity to maintain adequate performance. The storage and processing of data on IoT devices are also challenging due to storage and bandwidth restrictions. Therefore, models must be adjusted to handle smaller datasets or preprocess data locally before sending it for further analysis. In summary, the challenge for researchers has been and will continue to be the creation of IDS that protect data while maintaining high efficiency.4.3 Adopted deep learning algorithms.

Computational and resource constraints

The usage of deep learning faces several obstacles. Accessing representative IoT datasets is difficult, which complicates the generalization of results. Moreover, the complexity of deep learning models sometimes makes it challenging to understand the decisions made by the system. Limitations in computing power and memory on IoT devices restrict the feasibility of certain models. Finally, the security of IoT data is a major concern, with risks of adversarial attacks aimed at manipulating data or circumventing the detection system.

CONCLUSION

The present paper demonstrates that developments of IDSs are important for keeping IoT networks more secure, especially, with the increasing number of attacks such as denial of service and data theft. Recently, the advancements in artificial intelligence demonstrate the significant role to enhance the IDS capability and performance to detect malicious activities. In particular, deep learning algorithms are highly effective with large amounts

of data, even as traffic on IoT networks continues to grow. In this, study we have reviewed the most important features that are related to blinding and IDS based on DL algorithms. Our studies focus on the recent published works. As we have demonstrated, the most part of those studies are based on CNN and LSTM to build IDS. The combination of those algorithms demonstrates that the accuracy and precision can be improved.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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